



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

QUANTIZATION PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A AI/ML

TOTAL: 10 QUESTIONS

Q1 What is Quantization and what AI/ML use case is it designed for?

Threat Model

Q6 How do you evaluate a model's performance in Quantization?

Observability

Q2 What are the core abstractions in Quantization?

Architecture

Q7 How do you deploy a model trained with Quantization?

Lifecycle

Q3 How does Quantization handle training or inference?

Configuration

Q8 How does Quantization handle distributed or multi-GPU training?

Compliance

Q4 How does Quantization manage data preprocessing?

Hardening

Q9 How do you track experiments and versions in Quantization?

Testing

Q5 How does Quantization support GPU/TPU acceleration?

SSO / IdP

Q10 What are common pitfalls when using Quantization in production?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Quantization is a framework or service

Mitigation

Quantization is a framework or service targeting a specific AI/ML domain training neural networks, building LLM pipelines, deploying models, or orchestrating agents. Understanding its scope helps you know when to use it vs alternatives.

A6 Evaluation uses a held-out validation set

Observability

Evaluation uses a held-out validation set and metrics (accuracy, F1, BLEU, perplexity) appropriate to the task. Monitor both training and validation metrics to detect overfitting. Use a test set only at the very end to report final results.

A2 Quantization organizes work around abstractions

Primitives

Quantization organizes work around abstractions like models, tensors, pipelines, agents, or embeddings. Learning these core types is the first step to using the framework effectively.

A7 Export the model to a portable

Automation

Export the model to a portable format (ONNX, TorchScript, SavedModel). Serve it via a model server (TorchServe, TF Serving, Triton) or embed it in an API endpoint. Monitor inference latency and drift in production.

A3 For training, Quantization defines a model

Hardening

For training, Quantization defines a model architecture, a loss function, and an optimizer. For inference, a trained model processes input data and returns predictions. Batch processing and hardware acceleration are key performance levers.

A8 Quantization provides data-parallel or

Governance

Quantization provides data-parallel or model-parallel strategies to split work across GPUs. Data parallelism replicates the model on each GPU and averages gradients. Model parallelism splits layers across devices for models too large for one GPU.

A4 Quantization provides data loaders, tokenizers, or

Gotchas

Quantization provides data loaders, tokenizers, or pipeline components that transform raw data into the tensor or embedding format the model expects. Proper preprocessing is critical for model correctness and training speed.

A9 Use an experiment tracker (MLflow, Weights

Assessment

Use an experiment tracker (MLflow, Weights & Biases, Comet) alongside Quantization to log hyperparameters, metrics, and artifacts. Track the code version and data version for full reproducibility.

A5 Quantization detects available accelerators and

Federation

Quantization detects available accelerators and moves computation to them. Specify the device explicitly (cuda, mps, tpu) to avoid accidentally running expensive operations on CPU. Mixed-precision (fp16/bf16) further reduces memory and increases throughput.

A10 Common issues include training-serving skew

Optimization

Common issues include training-serving skew (different preprocessing), missing input validation, no latency SLA enforcement, models not versioned, and no process to retrain on drifted data distributions.